

	Assessment:	Project Milestone 3
	Subject:	Business Intelligence 381
	Total:	50

Project Milestone 3: Modelling + Evaluation + Deployment

Socioeconomic Development and Household Well-being in South Africa (SDHSA)

Purpose

Milestone 3 integrates three phases of the CRISP-DM methodology – Data Modelling, Evaluation and Deployment. Your team must build suitable analytical models, evaluate them technically and against the original business objectives, select a final approach and make the result usable through a deployed R-based analytical component.

Key principle

A high-performing model is not automatically the best model. Selection must consider the business problem, data characteristics, interpretability, fairness, limitations and deployment context.

Tasks

This milestone consists of the following steps that should be submitted as an assignment on given due date.

Task 1 – Modelling Strategy

- Restate the analytical question and identify the modelling task: classification, regression, clustering/segmentation, or another lecturer-approved technique.
- Select at least two defensible candidate modelling approaches where comparison is meaningful.
- Explain assumptions, parameter choices and why the techniques suit the data and business objective.
- Build models in R using the validated Milestone 2 analytical dataset and preprocessing pipeline. Preserve the Milestone 2 validation design and ensure that any learned preprocessing is fitted without information leakage from validation/test data.
- Use appropriate tuning or resampling where justified.
- Document the modelling techniques, model parameters, assumptions, and procedures for training, testing, and evaluating the models.

Task 2 – Technical Evaluation

- Use metrics appropriate to the modelling task; do not rely on one metric without justification.
- For classification, consider class distribution, confusion matrix and suitable metrics such as precision, recall, F1, specificity, ROC-AUC or balanced accuracy where appropriate.
- For regression, use suitable error and goodness-of-fit measures and inspect residual behaviour where relevant.

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- For clustering, evaluate cluster quality, stability and interpretability rather than treating cluster labels as ground truth.
- Compare candidate models using a clear validation matrix and select the final model with evidence.

Task 3 – Business and Ethical Evaluation

- Assess whether the results meet the business objectives and data-mining success criteria established in Milestone 1.
- Interpret key drivers/patterns carefully and distinguish association from causation.
- Identify limitations, uncertainty, possible bias and groups for whom the model may perform differently.
- Determine whether the result is sufficiently useful and responsible to deploy.
- Document all ethical issues that might arise from the tasks in this milestone and identify appropriate mitigation strategies.

Task 4 – Deployment

- Design a deployment strategy showing intended users, decisions supported, inputs, outputs and operational workflow.
- Evaluate the different deployment tools and motivate your recommended deployment tool.
- Deploy the selected model or analytical component in a user-friendly interface using the preferred deployment tool (e.g. R/Shiny or another lecturer-approved R-based interface)
- Document how new data would be validated and scored/processed.
- Define monitoring strategy and measures, performance thresholds, drift/change indicators and conditions requiring retraining or retirement.
- Keep deployment code in a clean, reproducible repository.

Required Deliverables

- Milestone 3 Modelling, Evaluation and Deployment Report (PDF).
- R Markdown/Quarto modelling notebook and supporting R scripts.
- Model comparison/evaluation matrix.
- Functional R/Shiny or approved R-based deployed analytical component.
- Clean code repository containing preparation, modelling and deployment code.
- Updated ethics and responsible AI usage log.

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Assessment Rubric

Criterion	Evidence expected	Marks
Modelling strategy and technique selection	Techniques align with question, data and assumptions.	8
Model implementation and reproducibility	Correct R implementation, tuning/resampling and reproducible code.	10
Technical model evaluation and comparison	Appropriate metrics, validation evidence and defensible model selection.	10
Interpretation and business evaluation	Results connected back to objectives, with limitations and non-causal interpretation.	8
Ethics, fairness and responsible AI evaluation	Bias/fairness/uncertainty risks are actively evaluated.	5
Deployment and monitoring strategy	Functional interface plus realistic monitoring, maintenance and retirement plan.	9
TOTAL:		50

Additional Information

- This is a group project. Please continue with your allocated project groups.
- All work must be original. Copying another group's work will not be tolerated.
- Includes names of all group members on the cover page.
- Submit your project milestones electronically on Moodle (BC Connect) before the due dates.
- All writing must be correctly cited and referenced, and the use of AI must be explicitly acknowledged.
- **Plagiarism is a serious offence.** Belgium Campus makes use of software that can scan for plagiarism and penalties apply for any plagiarised work including getting a **0** for the assignment.
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late assignments will not be accepted; missing the deadline is an automatic **0**.